Prajwal Khanal

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ITC, University of Twente, Enschede, the Netherlands

Research Interests

Remote sensing of vegetation; soil–vegetation–atmosphere interaction; land surface models and deep learning.

Education

PhD, Geo-Information Science and Earth Observation, *University of Twente* — *Feb 2023—. . .*
- WUNDER: field-scale soil moisture and fluxes for agricultural drought monitoring.
- Supervisors: Prof. dr. Bob Su and dr. Yijian Zeng.

M.Sc., Environmental Engineering, *Technical University of Munich* — *Nov 2020—Dec 2022*
- Graduated 1.4/5.0. Thesis: soil moisture controls on vegetation, in *Biogeosciences*.

B.E., Civil Engineering, *Institute of Engineering, Tribhuvan University, Nepal* — *Nov 2014—Aug 2018*
- Ranked 20/192, 77.7%. Thesis: flood inundation mapping, Babai River.

Employment

Research Intern, *Max Planck Institute for Biogeochemistry, Jena* — *Nov 2020—Feb 2022*
- Near-surface soil moisture vs. terrestrial water storage, controls on global vegetation.

Student Research Assistant, *Technical University of Munich* — *Nov 2021—Mar 2022*
- Flood inundation maps for the Ouémé (Benin) and Naryn (Kyrgyzstan) rivers, HEC-RAS.

Working Student, *Jena-Geos Ingenieurbüro GmbH, Munich* — *May 2021—Dec 2021*
- Excel/VBA tool for greenhouse-gas accounting: combustion, refrigeration, transport.

Civil Engineer (freelance), *ECEP, Kathmandu, Nepal* — *Dec 2018—Dec 2022*
- Business plans for community water supply projects, Government of Nepal.
- Smart water supply master plan for Lumbini, Ministry of Urban Development.

Publications

Peer-reviewed

Moutahir, H., Sulzer, M., Kiese, R., . . . **Khanal, P.**, et al. (2026). Matching scales of eddy covariance measurements and process-based modelling: assessing spatiotemporal dynamics of carbon and water fluxes in a mixed forest in Southern Germany. *Biogeosciences*, 23, 1719–1738.

Zeng, Y., Alidoost, F., Schilperoort, B., . . . **Khanal, P.**, et al. (2025). Towards an open soil-plant digital twin based on the STEMMUS-SCOPE model, following open science. *Computers & Geosciences*, 205, 106013.

Khanal, P., Hoek van Dijke, A. J., Schaffhauser, T., et al. (2024). Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning. *Biogeosciences*, 21(6), 1533–1547.

Bhattarai, P., **Khanal, P.**, Tiwari, P., et al. (2019). Flood inundation mapping of Babai basin using HEC-RAS and GIS. *Journal of the Institute of Engineering*, 15(2), 32–44.

Under review

Khanal, P., Zeng, Y., Daoud, M. G., Beckers, J., Su, Z. Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model. Preprint on SSRN.

Khanal, P., et al. Deep learning soil moisture and fluxes with multimodal foundation-model embeddings.

Conference contributions

Khanal, P., et al. (2026). Bridging the scale gap: daily 10-metre soil moisture, evapotranspiration and carbon fluxes via multimodal deep learning. *Workshop on Machine Learning for Land and Hydrology*, ECMWF, Reading, UK.

Khanal, P., Hoek van Dijke, A., Zeng, Y., Orth, R. (2023). Near-surface vs. sub-surface soil moisture impacts on vegetation functioning. *EGU General Assembly*, Vienna, EGU23-7670.

Khanal, P. (2019). Flood inundation mapping: an inference on the outbreak of flood-borne diseases. *International Conference on Health Engineering in Disaster*.

Awards

Second place, Hackathon on Machine Learning for Earth System Modelling — *Aug 2026*
- Quantization in foundation models. CESOC, ECMWF, TRA Modelling, FZ Jülich, Cologne.

Scholarships

Deutschlandstipendium, national merit scholarship, *TU Munich* — *2021/22*

Merit-based scholarship, *Institute of Engineering, Tribhuvan University* — *2014—2018*

Teaching

Teaching Assistant, *Technical University of Munich* — *2021—2022*
- River Basin Modelling; Integrated Water Resources Management; Rapidly Varying Flow.

Assistant Lecturer, *Cosmos College of Management and Technology, Nepal* — *Nov 2018—Jan 2021*
- Elective: Python and GIS for water resources.

Software

WUNDER data viewer, *ITC, University of Twente* — *2026*
- Weather and soil moisture loggers, read by growers at the food forests as drought develops. Source.

Skills

Modelling: STEMMUS-SCOPE, land surface models, HEC-RAS, SWAT
Programming: Python, PyTorch, R, GIS, Excel/VBA
Languages: English (C2), German (A1), Dutch (A2), Hindi, Nepali